

1.) Write general expressions for the electric and magnetic fields of a 1 GHz sinusoidal plane wave traveling in the +y direction in a lossless non-magnetic medium with relative permittivity $\epsilon_r = 9$. The electric field is polarized along the x direction, its peak value is 6 V/m, and its intensity is 4 V/m at $t = 0$ and $y = 2$ cm. (10 points)

General

$$E(y, t) = E_0 \cos(\omega t - k\hat{k} \cdot \mathbf{r} + \phi_0) \hat{e}$$

$$E(y, t) = 6 \cos(2\pi \times 10^9 t - k_y + \phi_0) \hat{x}$$

$$E(0.02, 0) = 6 \cos(2\pi \times 10^9(0) - 20\pi(0.02) + \phi_0) \hat{x}$$

$$4 = 6 \cos(0 - 0.4\pi + \phi_0) \hat{x}$$

$$4/6 = \cos(-0.4\pi + \phi_0)$$

$$\cos^{-1}(2/3) = -0.4\pi + \phi_0$$

$$E(y, t) \phi = \cos^{-1}(2/3) + 0.4\pi$$

$$= 2.1 \text{ rad}$$

General

$$\omega = 1 \times 10^9 \pi 2 = 2\pi \times 10^9 \text{ Hz}$$

$$\hat{k} \cdot \mathbf{r} = \hat{y} \cdot (x\hat{x} + y\hat{y} + z\hat{z}) = y$$

$$E_0 = 6 \text{ V/m}$$

$$E(0, 0.02, 0) = 4 \text{ V/m}$$

$$\hat{e} = \hat{x}$$

$$k = \frac{\omega \sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^9 \sqrt{9}}{3 \times 10^8} = 20\pi$$

Solution:

$$E(y, t) = 6 \cos(2\pi \times 10^9 t - 20\pi y + 2.1) \hat{x} \text{ V/m}$$

$$H = \frac{1}{\eta} (\hat{k} \times E)$$

$$= \frac{1}{\eta} (\hat{y} \times 6 \cos(2\pi \times 10^9 t - 20\pi y + 2.1) \hat{x})$$

$$= \frac{6}{125.67} \cos(2\pi \times 10^9 t - 20\pi y + 2.1) - \hat{z}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{9}} = 125.67$$

$$H = -47.7 \times 10^{-3} \cos(2\pi \times 10^9 t - 20\pi y + 2.1) \hat{z} \text{ A/m}$$

$$\tilde{E} = \frac{1}{\eta} \hat{k} \times \tilde{E}$$

$$E(y, t) = \frac{1}{\eta} (2\pi \times 10^9 t - 20\pi y + 2.1) \hat{x} \text{ V/m}$$

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2.) The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$E = 20 \cos((6\pi \times 10^9)t - kz) \hat{y} \text{ (V/m)}$$

$$\mu = \mu_0$$

$$\mu_r = 1$$

$$\rho_0 = 0$$

(12 points total)

Determine

(a) f , u_p , λ , k , and η . (5 points)

$$f = \frac{\omega}{2\pi} = \frac{6\pi \times 10^9}{2\pi} = 3 \times 10^9 \text{ Hz}$$

$$u_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8 \text{ m/s}}{\sqrt{2.56}} = 187.5 \times 10^6$$

$$\lambda = \frac{u_p}{f} = \frac{187.5 \times 10^6}{3 \times 10^9} = 62.5 \times 10^{-3} \text{ m}$$

$$k = \frac{\omega}{u_p} = \frac{6\pi \times 10^9}{187.5 \times 10^6} = 101 \text{ rad/m}$$

$$\text{Next time use } k = \frac{\omega \sqrt{\epsilon_r}}{c}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377 \Omega}{\sqrt{2.56}} = 235.6 \Omega$$

(b) the magnetic field H . (7 points)

$$H = \frac{1}{\eta} (\hat{k} \times E)$$

$$= \frac{1}{235.6} (\hat{z} \times 20 \cos(\omega t - kz) \hat{y})$$

$$= \frac{1}{235.6} (-\hat{x} 20 \cos(\omega t - kz))$$

$$H = -\hat{x} 84.8 \cos(6\pi \times 10^9 t - 101 z) \text{ mA/m}$$

$\hat{z} \times \hat{y} = -\hat{x}$ b/w, no need to do det

$$E = 20 \cos(\omega t - kz) \hat{y}$$

$$\hat{k} = \hat{z}$$

$$\eta_0 = 377 \Omega, \epsilon_r = 2.56$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = 235.6 \Omega$$

$$\hat{z} \times 20 \cos(\omega t - kz) \hat{y} = -\hat{x} 20 \cos(\omega t - kz)$$

$$\begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ 0 & 20 \cos(\omega t - kz) & 0 \end{bmatrix}$$

$$\hat{x} [0 - 20 \cos(\omega t - kz)] - \hat{y} [0 - 0] + \hat{z} [0 - 0]$$

$$= -\hat{x} 20 \cos(\omega t - kz)$$

Because it only care about direction

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3.) For a wave characterized by the electric field

$$\mathbf{E}(z, t) = a_x \cos(\omega t - kz) \hat{x} + a_y \cos(\omega t - kz + \delta) \hat{y},$$

identify the polarization state, determine the polarization angles (γ, χ), and sketch the locus of $\mathbf{E}(0, t)$ for each of the following cases.

(16 points total)

$$\mathbf{E}(0, t) = a_x \cos(\omega t) \hat{x} + a_y \cos(\omega t + \delta) \hat{y}$$

(a) $a_x = 3 \text{ V/m}$, $a_y = 4 \text{ V/m}$, and $\delta = 0$ radians (4 points)

$$\mathbf{E}(0, t) = 3 \cos(\omega t) \hat{x} + 4 \cos(\omega t) \hat{y}$$

Since $\delta = 0$, polarization is linear

(ψ_0) Auxiliary angle

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 0.9273 \text{ rads}$$

(γ) Rotation angle

$$\tan(2\gamma) = \frac{\tan(2\psi_0) \cos \delta}{1 - \tan^2 \psi_0 \cos \delta}$$

$$\tan(2\gamma) = \frac{\tan(2(0.9273)) \cos(0)}{1 - \tan^2(0.9273) \cos(0)}$$

$$\tan(2\gamma) = -3.428$$

$$2\gamma = \tan^{-1}(-3.428)$$

$$\gamma = -0.6435 \text{ rad but must be same sign as } \cos(\delta)$$

(b) $a_x = 3 \text{ V/m}$, $a_y = 4 \text{ V/m}$, and $\delta = \pi$ radians (4 points)

Since $\delta = \pi$, polarization is linear also +

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = 0.9273 \text{ rads}$$

$$\tan(2\gamma) = \frac{\tan(2\psi_0) \cos \delta}{1 - \tan^2 \psi_0 \cos \delta}$$

$$\tan(2\gamma) = \frac{\tan(2(0.9273)) \cos \pi}{1 - \tan^2(0.9273) \cos \pi}$$

$$\tan(2\gamma) = 3.428$$

$$2\gamma = \tan^{-1}(3.428)$$

$$\gamma = 0.6435 \text{ rads, but } \cos(\delta) = -1$$

to match the sign I need to $-\frac{\pi}{2}$

$$\gamma_{\text{new}} = 0.6435 - \frac{\pi}{2}$$

$$\gamma_{\text{new}} = -0.9273 \text{ rads matches sign of } \cos(\delta)$$

$$\sin(2\chi) = \sin(2\psi_0) \sin(\delta)$$

$$\sin(2\chi) = \sin(2 \times 0.9273) \sin(\pi)$$

$$\chi = 0$$

$\chi = 0$ makes sense, $\chi_{\text{aux}} = 0$, line, linear polarization

$$\mathbf{E}(0, t) = 3 \cos(\omega t) \hat{x} + 4 \cos(\omega t + \pi) \hat{y}$$

$$\text{let } \omega t = 0, \pi, \pi/2$$

$$(3, -4), (-3, 4), (0, 0)$$

$$\text{so, } \gamma_{\text{new}} = -0.6435 + \frac{\pi}{2} = 0.9273 \text{ rad}$$

Find χ Ellipticity angle (χ)

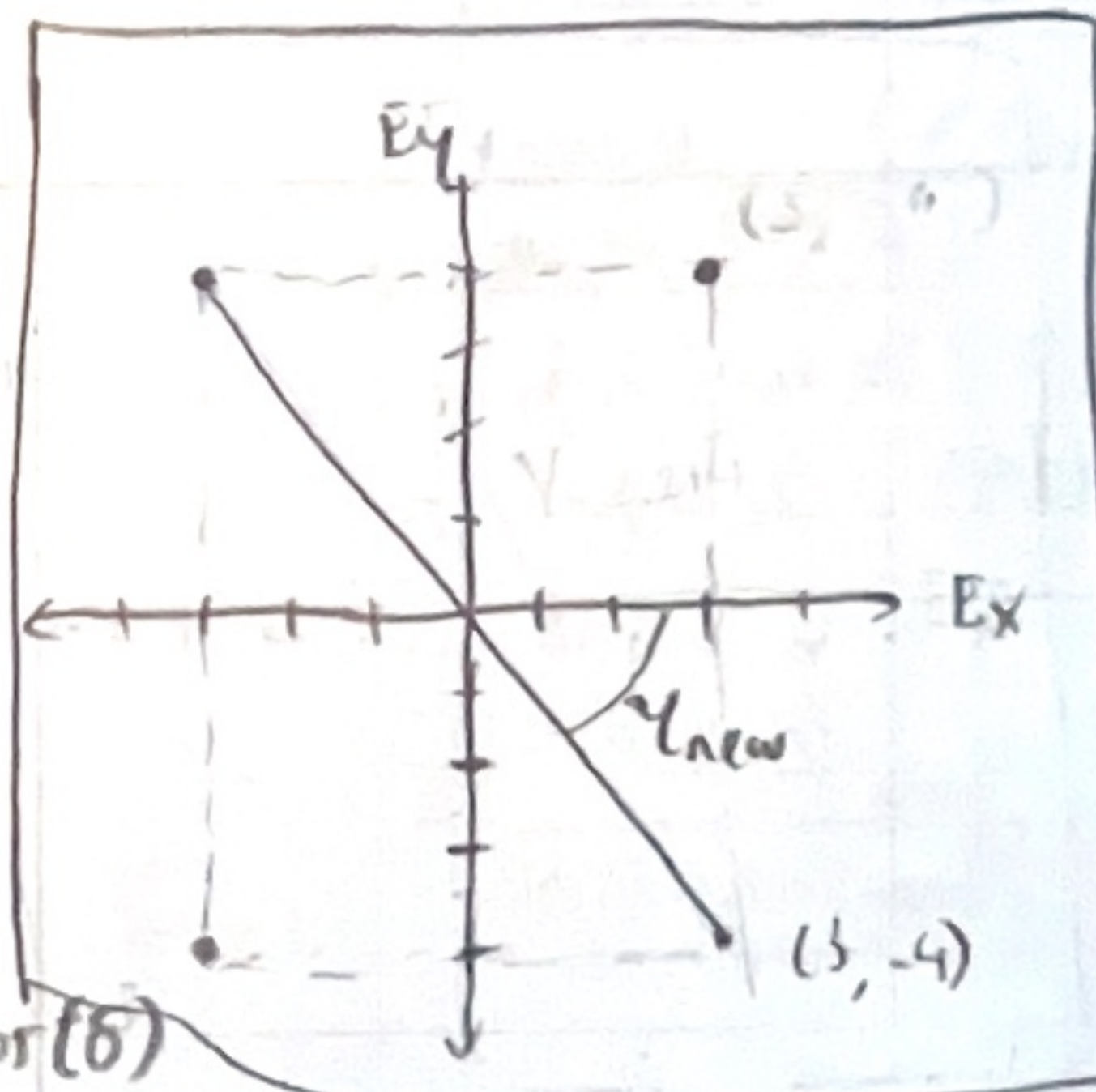
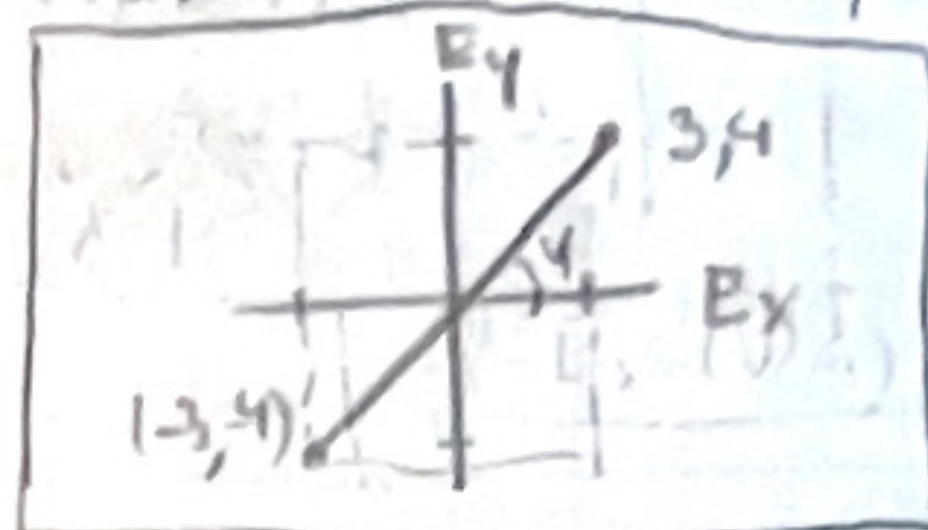
$$\sin(2\chi) = (\sin(2\psi_0)) \sin(\delta)$$

$$\sin(2\chi) = \sin(2(0.9273)) \sin(0)$$

$$\sin(2\chi) = 0$$

$\chi = 0$ rad, and that's why it's linear

$$\mathbf{E}(0, t) = 3 \cos(\omega t) \hat{x} + 4 \cos(\omega t) \hat{y}$$



ask mark

(c) $a_x = 3 \text{ V/m}$, $a_y = 3 \text{ V/m}$, and $\delta = \pi/4$ radians (4 points)

Aux angle (PSI)
Ellipticity Angle (chi)

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = 0.7853 \text{ rads}$$

$$\sin 2\chi = \sin(2\psi_0) \sin \delta$$

$$\sin 2\chi = \frac{\sqrt{2}}{2}$$

$$\chi = 0.3927 \text{ rad}$$

$\sin \chi > 0$, it's LHEP $\sim$

$$\tan(2\chi) = \tan(2\psi_0) \cos(\delta)$$

$$\tan(2\chi) = -141.1 \times 10^9$$

$$\chi = -0.7854, \cos(\frac{\pi}{4}) \text{ is not } -50.$$

$$\chi_{\text{new}} = -0.7854 + \frac{\pi}{2}$$

$$\chi_{\text{new}} = 0.7854 \text{ rad} \sim 45^\circ$$

Rotation angle (gamma)

Axial Ratio (R)
"Factor of EP"

$$R = \frac{1}{\tan|\chi|} = \frac{1}{\tan(0.3927)} = 2.4$$

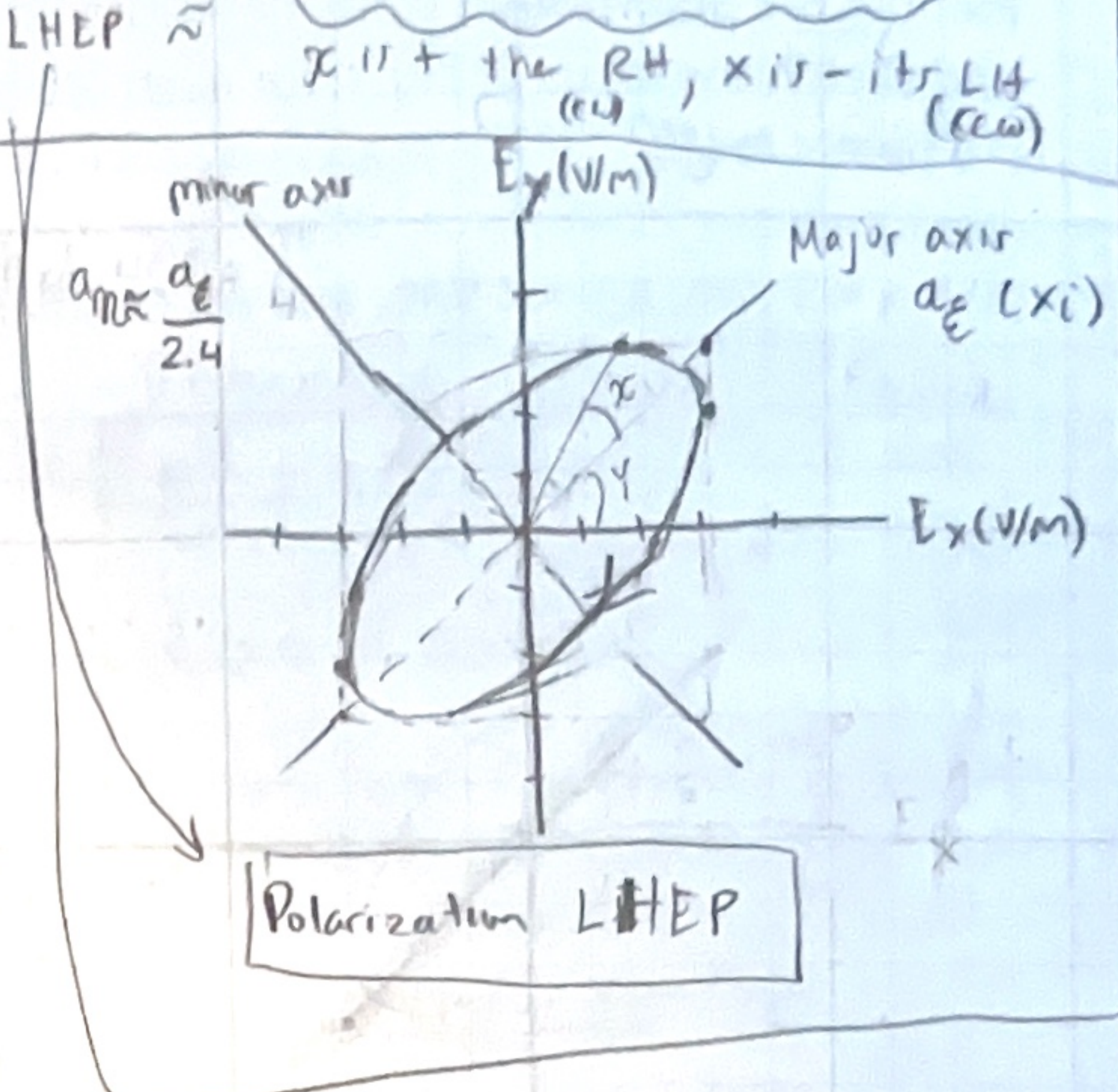
so major axis is $2.4 \times$ larger than minor axis

$$E(0, t) = 3 \cos(\omega t) \hat{x} + 3 \cos(\omega t + \delta) \hat{y}$$

$$\text{Let } \omega t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$$

$$(3, 2.12), (0, 0), (-3, -2.12)$$

$\chi = 0$ means linear
 $\chi = \frac{\pi}{4}$ means circle
 $0 < \chi < \frac{\pi}{4}$ means ellipse



(d) $a_x = 3 \text{ V/m}$, $a_y = 4 \text{ V/m}$, and $\delta = -3\pi/4$ radians (4 points)

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = 0.9273$$

$$\sin(2\chi) = \sin(2\psi_0) \sin \delta$$

$$\sin(2\chi) = -0.6788$$

$$\chi = -0.3731 \text{ rads}$$

RHEP Polarization ($\chi < 0$)

$$\tan(2\chi) = \tan(2\psi_0) \cos(\delta)$$

$$\tan(2\chi) = 2.424$$

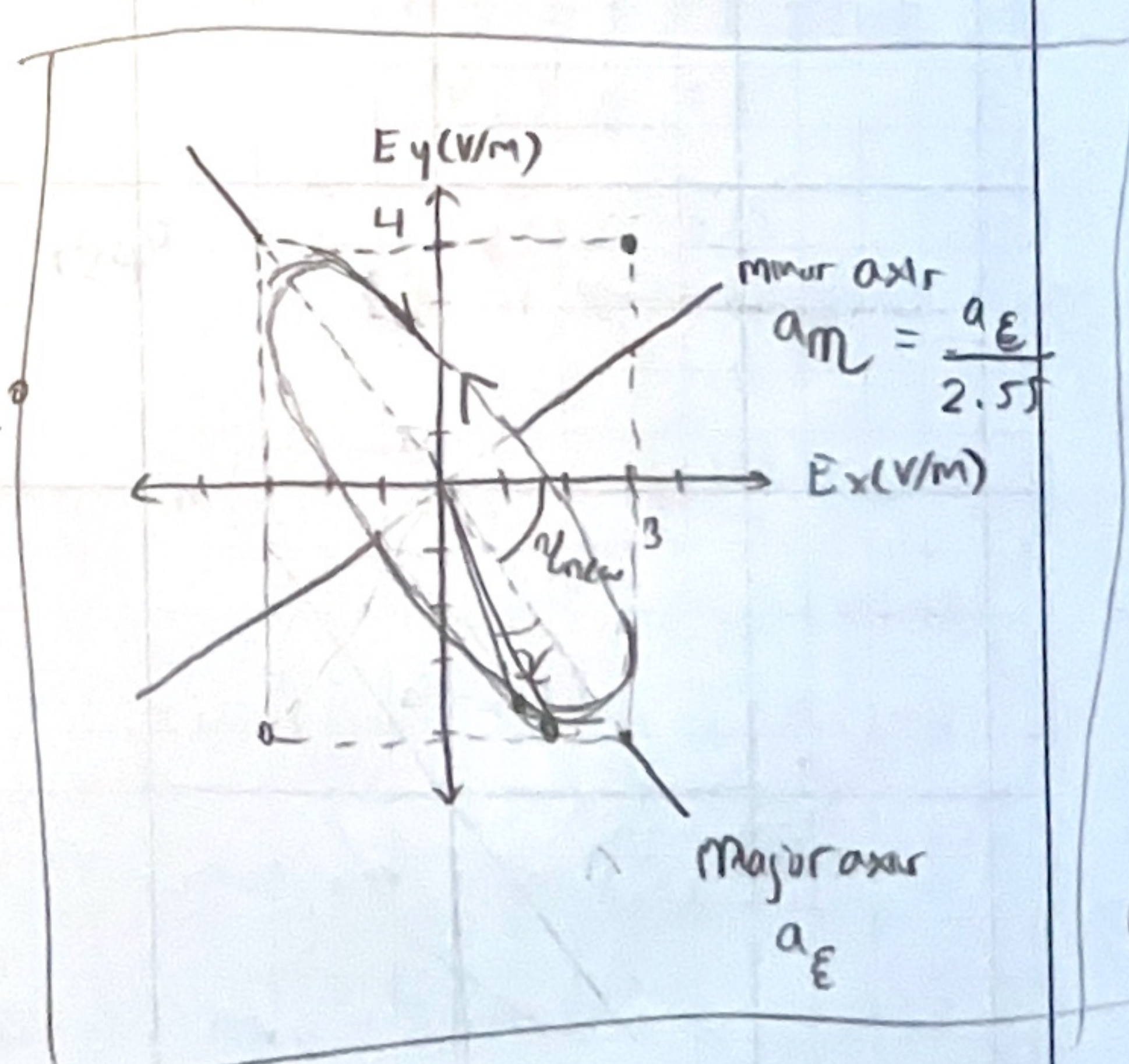
$$\chi = 0.5898, \text{ but } \cos \delta = -50,$$

$$\chi_{\text{new}} = (0.5898 - \frac{\pi}{2})$$

$$\chi_{\text{new}} = -0.9810 \approx \chi_{\text{new}} \frac{180}{\pi} \approx -56.2^\circ$$

$$R = \frac{1}{\tan|\chi|} = \frac{1}{\tan(0.3731)} = 2.55 \times \text{larger than minor axis}$$

$$E(0, t) = 3 \cos(\omega t) \hat{x} + 4 \cos(\omega t + \delta) \hat{y}$$



4. A linearly polarized plane wave of the form

$$\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{\mathbf{x}}$$

can be expressed as the sum of an RHC-polarized wave with magnitude a_R and an LHC-polarized wave with magnitude a_L . Prove this statement by finding expressions for a_R and a_L in terms of a_x . (8 points)

$$\text{RHC: } \tilde{\mathbf{E}}_R = a_R (\hat{\mathbf{x}} - j\hat{\mathbf{y}}) e^{-jkz}$$

$$\text{LHC: } \tilde{\mathbf{E}}_L = a_L (\hat{\mathbf{x}} + j\hat{\mathbf{y}}) e^{-jkz}$$

$$\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L = [(a_R + a_L)\hat{\mathbf{x}} + j(-a_R + a_L)\hat{\mathbf{y}}] e^{-jkz}$$

Match $\tilde{\mathbf{E}}$

$$\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{\mathbf{x}}$$

$$\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{\mathbf{x}} + 0\hat{\mathbf{y}}$$

$$\tilde{\mathbf{E}} = [(a_R + a_L)\hat{\mathbf{x}} + j(-a_R + a_L)\hat{\mathbf{y}}] e^{-jkz}$$

$$a_x = a_R + a_L$$

must be equal:

$$a_x = 2a_R \quad \text{or} \quad a_x = 2a_L$$

$$a_R = \frac{a_x}{2}$$

$$a_L = \frac{a_x}{2}$$

$$\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L = a_x e^{-jkz} \hat{\mathbf{x}} = \tilde{\mathbf{E}}$$

$$a_R = a_L = \frac{a_x}{2}$$

Prove it

$$\begin{aligned} \tilde{\mathbf{E}} &= \tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L = [(a_R + a_L)\hat{\mathbf{x}} + j(-a_R + a_L)\hat{\mathbf{y}}] e^{-jkz} \\ &= a_x e^{-jkz} \hat{\mathbf{x}} \quad \text{QED} \\ &= \tilde{\mathbf{E}} \end{aligned}$$

Dry soil so $\epsilon_r = 1$

5. Dry soil is characterized by $\epsilon_r = 2.5$, $\mu_r = 1$, and $\sigma = 10^{-4} \text{ S/m}$. At each of the following frequencies, determine if dry soil may be considered a good conductor, a quasi-conductor, or a low-loss dielectric. Then calculate α , β , λ , u_p , and η_c . (24 points total)

(a) 60 Hz (6 points)

$$\omega = 2\pi f = 2\pi 60 = 120\pi, \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

Loss tangent

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0} = \frac{(10^{-4})}{120\pi(2.5)(8.85 \times 10^{-12})} = 119.91 \times 10^3 > 100 \text{ so good conductor}$$

$$\alpha \approx \beta \approx \sqrt{\pi f \mu_0 \sigma} = \sqrt{\pi 60 (4\pi \times 10^{-7} \text{ H/m}) 10^{-4}} = 1.54 \times 10^{-6} \text{ Np/m and rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{1.54 \times 10^{-6}} = 40.8 \times 10^3 \text{ m}$$

$$u_p = \sqrt{\frac{4\pi f}{\mu_0 \sigma}} = \sqrt{\frac{4\pi 60}{(4\pi \times 10^{-7}) 10^{-4}}} = 2.45 \times 10^6 \text{ m/s}$$

$$\eta_c = (1+j) \frac{\alpha}{\sigma} = (1+j) \frac{1.54 \times 10^{-6}}{10^{-4}} = (1+j)(1.54) \Omega \quad \angle 45^\circ, 1.54(1+j) \Omega$$

(b) 1 kHz (6 points)

$$\omega = 2\pi f = 2\pi(1000) = 2000\pi$$

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0} = \frac{10^{-4}}{2000\pi(2.5)(8.85 \times 10^{-12})} = 719 > 100, \text{ good conductor}$$

$$\alpha \approx \beta \approx \sqrt{\pi f \mu_0 \sigma} = \sqrt{\pi(1000)(4\pi \times 10^{-7}) 10^{-4}} = 6.28 \times 10^{-6} \text{ Np/m and rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{6.28 \times 10^{-6}} = 10 \times 10^3 \text{ m}$$

$$u_p = \sqrt{\frac{4\pi f}{\mu_0 \sigma}} = \sqrt{\frac{4\pi(1\text{K})}{4\pi \times 10^{-7} (10^{-4})}} = 10 \times 10^6 \text{ m/s}$$

$$\eta_c = (1+j) \frac{\alpha}{\sigma} = (1+j) \frac{6.28 \times 10^{-6}}{10^{-4}} = 6.28(1+j) \Omega$$

$$\mu = \mu_r \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

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(c) 1 MHz (6 points) $\delta = 10^{-4}$, $\omega = 2\pi(1 \times 10^6) = 6.28 \times 10^6 \text{ rad/s}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $\epsilon_r = 2.5$

$$\frac{\epsilon''}{\epsilon'} = \frac{\delta}{\omega \epsilon_r \epsilon_0} = \frac{10^{-4}}{(6.28 \times 10^6)(2.5)(8.85 \times 10^{-12})} = 7.19 \times 10^{-3} < 0.01 < \frac{\epsilon''}{\epsilon'} < 100 \text{ Quasi-conductor}$$

$$X = \sqrt{1 + \left(\frac{\delta}{\omega \epsilon}\right)^2} = \sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} = \sqrt{1 + (7.19 \times 10^{-3})^2} = 1.232$$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon_0 \epsilon_r}{2}} \sqrt{X-1} = (6.28 \times 10^6) \sqrt{\frac{(4\pi \times 10^{-7})(8.85 \times 10^{-12})(2.5)}{2}} \sqrt{1.232-1} = 11.27 \times 10^{-3} \text{ NP/m}$$

$$\beta = \omega \sqrt{\frac{\mu \epsilon_0 \epsilon_r}{2}} \sqrt{X+1} = 23.4 \times 10^{-3} \sqrt{1.232+1} = 34.98 \times 10^{-3} \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{34.98 \times 10^{-3}} = 179.6 \text{ m}$$

$$\mu_p = \frac{\omega}{\beta} = \frac{6.28 \times 10^6}{34.98 \times 10^{-3}} = 179.5 \times 10^6 \text{ m/s}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{2.5}} = 238.4 \Omega$$

$$|\eta_c| = \frac{\eta}{X^{1/2}} = \frac{238.4}{\sqrt{1.232}} = 214.8 \Omega \quad \theta_n = \frac{1}{2} \tan^{-1}(0.719) \approx 17.9^\circ$$

$$\eta_c = 214.8 \angle 17.9^\circ \Omega$$

$$\text{Real: } 214.8 \cos(17.9) = 204.4$$

$$\text{Imag: } 214.8 \sin(17.9) = 66.1$$

$$204.4 + j66.1 = \eta_c$$

(d) 1 GHz (6 points) $\delta = 10^{-4}$, $\omega = 2\pi(1 \times 10^9) = 2\pi \times 10^9$, $c = 3 \times 10^8 \text{ m/s}$

$$\frac{\epsilon''}{\epsilon'} = \frac{\delta}{\omega \epsilon_r \epsilon_0} = \frac{10^{-4}}{(2\pi \times 10^9)(2.5)(8.85 \times 10^{-12})} = 7.19 \times 10^{-6} < 10^{-2}, \text{ Low loss dielectric}$$

$$\alpha = \frac{\delta \omega}{2}, \quad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377 \Omega}{\sqrt{2.5}} = 238.4 \Omega, \quad \alpha = \frac{10^{-4} \cdot 238.4}{\sqrt{2}} = 11.9 \times 10^{-3} \text{ NP/m}$$

$$\beta = \frac{\omega \sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^9 \sqrt{2.5}}{3 \times 10^8} = 33.1 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{33.1} = 189.7 \times 10^{-3} \text{ m}$$

$$\mu_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.5}} = 189.7 \times 10^6 \text{ m/s}$$

$$\eta_c = \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{2.5}} = 238.4 \Omega$$

Fix me ✓

Let + 12-28
14-16 23-27

6. In a nonmagnetic, lossy, dielectric medium, a 300 MHz plane wave is characterized by the magnetic field phasor

$$\tilde{\mathbf{H}} = (\hat{x} - j4\hat{z})e^{-2y}e^{-j9y} \quad (\text{A/m})$$

Given: $f = 300 \text{ MHz} = 300 \times 10^6 \text{ Hz}$, $\omega = 2\pi f = 600\pi \times 10^6 \text{ rad/s}$, propagation dir = y , $\alpha = 2 \text{ Np/m}$, $\beta = 9 \text{ rad/m}$

Obtain time-domain expressions for the electric and magnetic field vectors. (10 points)

Propagation direction: $+y$ because $e^{-\alpha y}e^{-j\beta y}$
 $\alpha = 2 \text{ Np/m}$, $\beta = 9 \text{ rad/m}$, $\omega = 2\pi(300 \times 10^6) = 6\pi \times 10^8 \text{ rad/s}$
 $\mu_r = 1$ because non-mag so $\mu = \mu_0 \mu_r = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

$$\begin{aligned} H(y,t) &= \text{Re} \{ \tilde{\mathbf{H}} e^{j\omega t} \} \\ &= \text{Re} \{ (\hat{x} - j4\hat{z}) e^{-2y} e^{-j9y} e^{j\omega t} \} \\ &= e^{-2y} \text{Re} \{ (\hat{x} + 4\hat{z} e^{-j\pi/2}) e^{-j9y} e^{j\omega t} \} \\ &= e^{-2y} \text{Re} \{ (\hat{x} + 4\hat{z} e^{-j\pi/2}) e^{j(\omega t - 9y)} \} \\ &= e^{-2y} \text{Re} \{ \hat{x} e^{j(\omega t - 9y)} + 4\hat{z} e^{j(\omega t - 9y - \pi/2)} \} \\ &= e^{-2y} (\hat{x} \cos(\omega t - 9y) + 4\hat{z} \sin(\omega t - 9y)) \end{aligned}$$

$$H(y,t) = e^{-2y} (\hat{x} \cos(6\pi \times 10^8 t - 9y) + 4\hat{z} \sin(6\pi \times 10^8 t - 9y)) \quad \text{A/m}$$

$$\begin{aligned} \tilde{\mathbf{E}}(y,t) &= \text{Re} [\tilde{\mathbf{E}} e^{j\omega t}] \\ \tilde{\mathbf{E}} &= -\mu_c (\hat{k} \times \tilde{\mathbf{H}}) \quad \mu_c = \frac{j\omega\mu}{\alpha + j\beta} = \frac{j236.9}{2 + j9} \quad \gamma = \alpha + j\beta \\ &= -256.9 e^{j0.2188} (\hat{y} \times \tilde{\mathbf{H}}) \quad j = e^{j\pi/2} \\ &= -256.9 e^{j0.2188} (-\hat{z} - 4\hat{x} e^{j\pi/2}) e^{-2y} e^{-j9y} = \frac{236.9 e^{j\pi/2}}{9.22 e^{j1.352}} \\ &= 256.9 e^{j(0.2188)} \Omega \quad 2 + j9 = \sqrt{2^2 + 9^2} \angle \tan^{-1}(\frac{9}{2}) = 9.22 e^{j1.352} \end{aligned}$$

$$\begin{aligned} \mathbf{E}(y,t) &= \text{Re} [\tilde{\mathbf{E}} e^{j\omega t}] \\ &= -256.9 e^{-2y} [-\hat{z} e^{j(\omega t - 9y + 0.2188)} + 4\hat{x} e^{j(\omega t - 9y + 0.2188 + \pi/2)}] \\ &= -256.9 e^{-2y} [-\hat{z} \cos(\omega t - 9y + 0.2188) + 4\hat{x} \cos(\omega t - 9y + 0.2188 + \pi/2)] \end{aligned}$$

$$\mathbf{E}(y,t) = -256.9 e^{-2y} [-\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) + 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

$$\mathbf{E}(y,t) = 256.9 e^{-2y} [\hat{z} \cos(6\pi \times 10^8 t - 9y + 0.2188) - 4\hat{x} \sin(6\pi \times 10^8 t - 9y + 0.2188)]$$

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$$\frac{\delta}{\delta} = 0 \text{ loss}$$

7. A wave traveling in a nonmagnetic medium with $\epsilon_r = 9$ is characterized by an electric field given by

$$\mu_r = 1, \text{ so } \mu = \mu_0 = 4\pi \times 10^{-7} \text{ H/m} \quad \hat{k} = -\hat{x}$$

$$\mathbf{E} = 3 \cos((\pi \times 10^7)t + kx) \hat{y} - 2 \cos((\pi \times 10^7)t + kx) \hat{z} \text{ (V/m)}.$$

$$\epsilon = \epsilon_r \epsilon_0 = 9 (8.854 \times 10^{-12}) = 79.69 \times 10^{-12}$$

$$\eta_0 = 377 \Omega$$

Determine the direction of wave travel and average power density carried by the wave. (10 points)

$$\text{Solve } S_{av} = \frac{|\mathbf{E}_0|^2}{2\eta} \hat{k}$$

$$|\mathbf{E}_0|^2 = 3^2 + (-2)^2 = 13$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \text{ or } \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{9}} = 125.67 \Omega$$

$$\hat{k} = -\hat{x} \text{ because } \omega t + (-x)k$$

$$S_{av} = \frac{|\mathbf{E}_0|^2}{2\eta} \hat{k}$$

$$= \frac{13}{2(125.67)} \hat{x}$$

$$= -51.2 \times 10^{-3} \hat{x} \text{ W/m}^2$$

Note sheet

$$\mathbf{E}(x, t) = E_1 \cos(\omega t \pm kx + \phi) \hat{u}_1 + E_2 \cos(\omega t \pm kx + \phi) \hat{u}_2$$

Base formula: $E = E_0 \cos(\omega t \pm k \hat{k} + \phi_0)$
Given then wave direction travel

$$(\omega t - kx) \text{ or } (-\omega t + kx) \quad \text{then } +\hat{x} \text{ travel}$$

$$(\omega t + kx) \text{ or } (-\omega t - kx) \quad -\hat{x}$$

Finding E_0

$$\mathbf{E} = 3 \cos(\dots) \hat{y} - 2 \cos(\dots) \hat{z}$$

$$\text{Factor out } E_0 = 3 \hat{y} - 2 \hat{z} \text{ amplitude vector}$$